

Siddhartha Reddy Munugala

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AI ENGINEER INTERN — APPLICATION

CS student with hands-on experience building and shipping AI-powered products — LLM pipelines, agentic workflows, RAG systems, and real-time full-stack applications. Comfortable working with incomplete information, iterating fast on real user feedback, and owning features end-to-end. Two live products, 1,621 combined users, and a track record of turning ambiguous problems into working systems.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL | Frameworks: Next.js 14, React, Node.js, FastAPI, Tailwind CSS
AI / LLM: Claude API (Anthropic), Gemini API, OpenAI API, LangChain, Whisper, RAG, prompt engineering, agentic workflows, embeddings
Infra: Supabase, PostgreSQL, Neon Postgres, Redis, Vercel, Render, Docker, Git, GitHub Actions
Other: WebSockets, WebRTC, REST APIs, FFmpeg, Demucs, XGBoost, SHAP, tree-sitter, NetworkX

EXPERIENCE

Co-Founder & Lead Engineer — VibeIn / ArenaHQ

Jan 2025 – Present

vibein.us · arenahq.us

- Built and shipped two full-stack AI-powered web apps (Next.js 14, Supabase, Gemini API, Twilio) serving 1,621 combined users with 20% active retention pre-beta.
- Integrated Gemini AI for real-time matchmaking logic and content moderation; resolved API deprecation mid-production and migrated models without downtime.
- Ran 50+ user interviews to turn customer problems into product changes; managed end-to-end deployment with 99%+ uptime.
- Resolved 8+ critical production incidents — auth failures, notification breakdowns, real-time failures — under hard deadlines.

IT Coordinator & Chapter Administrator — NSLS Executive Board

Sep 2024 – Present

University of South Dakota

- Built automated member tracking, onboarding workflows, and internal dashboards using Python and Google Apps Script; reduced onboarding latency by 50%.

Production Operations Staff — USD Athletics

Aug 2025 – Present

University of South Dakota · Part-Time On-Campus

- Operate live-event audio, display systems, and broadcast cues for NCAA events under zero-tolerance real-time constraints.

PROJECTS

EKO — AI Background Music & Voice Dubbing System

2026 – Present

- Built three AI-powered modes: Studio (vocal sketch → 10 BGM variations via Google Lyria 3), BGM Dub (scene-energy-matched music generation), and Voice Dub (multilingual speech replacement via Demucs stem separation + Whisper transcription).
- Designed and debugged full AI pipeline: segmentation, transcription, translation, synthesis, and audio reassembly — resolved critical memory constraint by migrating from local MusicGen-small (1.8GB) to HuggingFace Inference API (~400MB).
- 821 users, 40% DAU at zero paid acquisition; arXiv preprint (cs.SD) in progress. Stack: Next.js 14, FastAPI, Neon Postgres, Render, Vercel Blob, faster-whisper, librosa, Gemini Flash.

COGN — Cognitive Integrity Layer for AI Codebases

2026

- Built an agentic CI/CD GitHub Action that analyzes PRs for "cognitive drift" — reasoning-pattern shifts introduced by AI-generated code — using three custom primitives (cogn.signature, cogn.transfer, cogn.integrity) and AST-based scoring against repo baselines.

PRISM — PR Risk Intelligence & Structural Modeling

2026

- ML pipeline predicting post-merge engineering risk from 45 features (AST via tree-sitter, dependency graphs via NetworkX, code churn); XGBoost + SHAP explainability; AUC ≥ 0.80, ΔAUC < 0.05 cross-repo.

ALASKA — Encrypted AI Assistant for Military Veterans

2026 · Top-10 Hackathon

- 48-hour build: AES-256-GCM encrypted LLM assistant with VA routing algorithm, crisis detection, and accessibility modes.

WTFcode — Natural Language Dev CLI

2026

- Multi-agent agentic developer tool: natural language → structured multi-step reasoning → code execution pipeline; reduced token overhead 90% via directory import-map compression.

EDUCATION

University of South Dakota

Vermillion, SD • Expected May 2027

B.S. Computer Science — Minor in Innovation & Entrepreneurship